

P8 — bahan baca sebelum Submit (8 Oktober 2026)

Untuk Pak Amien. Berkas PDF ini memuat semua yang perlu dibaca sebelum Submit, berurutan: (1) halaman ini; (2) surat pengantar v0.4, 2 halaman, bertanda tangan, untuk editor; (3) tanggapan untuk reviewer v0.4, tanpa nama, bisa diteruskan editor ke reviewer; (4) Gambar 1, 3 dan 4 yang digambar ulang. Naskah lengkapnya tidak diulang di sini karena sudah Anda baca dan setuju pada 7 Oktober; yang berubah sejak itu hanya yang tercantum di bawah.

1. Yang berubah di naskah sejak 7 Oktober (hanya ini)

1. **Section 2.4, kalimat pertama (baru).** *A classifier is a statistical procedure that is given forms whose class is known, here coded or uncoded, learns which combinations of the inputs go with each class, and then gives any form a score for belonging to one of them.* — Reviewer 2 (butir 5) meminta konsep *classifier* dijelaskan untuk pembaca non-teknis.
2. **Section 2.4, sesudah “...shown in figure 1.” (baru).** *An arrow beside a bar in that figure gives the direction of the input, that is, the sign of the correlation over all forms between the input’s value and its attribution; a dot is printed where that correlation is below 0.3 in absolute value.* — Aturan di balik panah Gambar 1 sebelumnya tidak tertulis di mana pun.
3. **Section 2.5, lanjutan kalimat “...and the precision, recall and F1 score of the uncoded class” (baru).** : *precision is the share of the forms scored 0.5 or more that are uncoded, recall is the share of the uncoded forms that are scored 0.5 or more, and the F1 score is the harmonic mean of the two, which is high only when both are.* — Reviewer 2 (butir 12) meminta pembaca dituntun membaca tabel.
4. **Tabel 8.** Dua sel selisih daftar-yang-ditahan: -0.019 menjadi -0.020 (teks naskah memang menulis 0.020; nilai tak-dibulatkan -0.0197). Satu nama baris: “Pre-glottalised” menjadi “Pre-glottalized”.
5. **Daftar rujukan.** Rentang halaman Lundberg dan Lee (2017) dihapus: tidak dapat dipastikan di sumbernya (ada dua rentang yang beredar), dan pemeriksaan rujukan 5 Oktober memang menyarankan menghapusnya.

Judul, abstrak, klaim, tabel lain dan semua angka lain **tidak berubah**. Naskah tetap 24 halaman.

Keadaan 8 Oktober sore: semua berkas di bawah ini **sudah diunggah ke draf R1 di portal** (belum disubmit), README di Zenodo sudah diganti dengan yang bersih (DOI tetap), dan repo sudah di-push ke GitHub. Kalau sesudah membaca Anda ingin mengubah sesuatu, berkasnya tinggal diganti lagi sebelum Submit.

2. Yang berubah di gambar

- **Gambar 1.** Legenda “more likely a candidate” menjadi “more likely uncoded” (“candidate” adalah istilah kerja internal; naskah hanya memakai *coded* dan *uncoded*); “ $(|r| < 0.3)$ ” dihapus dari legenda dan aturannya ditulis di Section 2.4.
- **Gambar 3.** Angka persen di segmen berarsir diberi latar putih (sebelumnya tertimpa garis arsir).
- **Gambar 4.** Legenda dipindah ke bawah sumbu (sebelumnya menutupi dua baris terbawah); kata “chance” dipindah ke atas garisnya.

- **Kelima berkas gambar (TIFF)** diberi pita putih di bagian atas. Portal jurnal menempelkan cap besar “Figure N” di atas isi gambar pada PDF gabungan yang dibaca reviewer (di Gambar 2 cap itu menutupi kotak pertama); dengan pita itu cap jatuh di area kosong. Surat pengantar menyebutnya dalam satu kalimat.

Nilai yang diplot tidak berubah; isi Gambar 2 dan 5 tidak disentuh.

3. Keputusan untuk Anda (default berlaku bila Anda tidak berkata apa-apa)

	Keputusan	Default
a	Tiga kalimat baru di naskah dan dua sel Tabel 8 (bagian 1)	terima
b	Tiga gambar yang digambar ulang (bagian 4 berkas ini)	terima
c	Dua surat sebagai ganti satu: pengantar bertanda tangan + tanggapan tanpa nama	ya
d	Kalimat surat pengantar “...not having heard otherwise, we have handled them within this revision...”	tetap, kecuali Sander menjawab sebelum Submit
e	Butir kecil ketiga di R1-2 (kami mengutip tahun 2000 dan 1996; laporan reviewer menulis 2001 dan 1989)	tetap; satu kalimat untuk dicoret bila tidak diinginkan
f	Pita putih di atas kelima berkas gambar dan kalimatnya di surat pengantar	terima
g	Rentang halaman satu rujukan (Lundberg dan Lee 2017) dihapus	terima

Bila Anda menolak (a) atau (b): jalur cadangan tertulis di PACKAGE_R1_20261008.md §2 — naskah dan gambar lama dipertahankan dan tiga tempat di surat tanggapan dilunakkan.

4. Yang perlu Anda perhatikan saat membaca surat

- Surat tanggapan menyebut hal-hal yang **kita akui keliru di naskah pertama** (Bagian C, 13 butir). Itu sama dengan isi catatan ke editor 7 Oktober; tidak ada pengakuan baru.
- Dua jawaban sengaja berkata “kami tidak punya penjelasan” atau “studi ini tidak dapat menunjukkannya” (R1-13 a tentang Tolaki; R2-21 tentang ‘One Hundred’). Itu pilihan sadar: lebih baik daripada jawaban yang mengelak.
- Di penutup Bagian B ada kalimat bahwa sebagian besar *Discussion* versi pertama — bagian yang disukai reviewer 2 — ditarik, beserta alasannya.

Cover letter — revised manuscript OL-03-2026-11

To the Editors of *Oceanic Linguistics*

Dear Professor Adelaar,

We submit the revised version of manuscript OL-03-2026-11, now titled *Uncoded basic vocabulary in six Sulawesi word lists of the Austronesian Basic Vocabulary Database*. We are grateful to both reviewers and to the editors. The first reviewer's comments corrected our classification and terminology and sent us to the literature on Makasar and Tolaki; the second reviewer's highlights told us, point by point, where a reader of the journal would be lost in Sections 2 and 3. We have tried to answer every one of them.

As we wrote in our note of 7 October 2026, the revision also corrects errors of our own that we found when we re-derived every number of the submitted text from the raw ABVD files. They change the abstract, the title and the claims of the paper: the classifier is now presented as a ranking aid rather than a detector of substrate vocabulary; the “two-method consensus”, the clustering, the ablation table and the section on Javanese script are removed, and the application to sixteen further lists is reduced to one paragraph without a geographic claim; the negative result about a shared layer is replaced by a statement of what a permutation test measures; and two subsections on Makasar and Tolaki answer the first reviewer's questions with counts from the data. Each correction is listed in Part C of the response to the reviewers. In that note we asked whether you would prefer another procedure for these corrections, such as a further look by the reviewers; not having heard otherwise, we have handled them within this revision, and we will follow whichever procedure you prefer.

The files

- **Article file.** Anonymised: author names and affiliations, the DOI of the data release and the address of the code repository are withheld in it. It is 24 pages in the journal's template, with eight tables. A version with names is available on request.
- **Figures.** Five, uploaded separately as TIFF files. Figure 1 replaces the plot whose caption the first reviewer gave as an example of terminology that cannot be followed; the other four are new, added for readers who do not work with classifiers: a workflow diagram, two charts that draw the numbers of Tables 2 and 4, and a chart of the written glottal mark by list. Each figure file carries a white band at the top, so that the figure label which the submission system prints on the merged PDF does not cover the drawing; the band can be cropped.
- **Response to the reviewers.** A Word file with bulleted answers, as the Managing Editor advised on 8 October 2026. We have kept one bullet for every point of the two reports, so that each reviewer finds every point, with more detail where a point is substantive, and the corrections of our own are in a separate part. It carries no author names, so that it can be passed to the reviewers.
- **Tracked changes.** A copy of the article with tracked changes is added as supplemental material; the article file is the clean copy. The first submission was a PDF, so the tracked copy compares a Word conversion of the submitted text with the revised article file; in the conversion, citations and references are set in the style of the revised text and images are omitted, so that the markup shows content rather than formatting. Because Sections 2 and 3 were largely rewritten and several sections removed, much of the markup consists of replaced blocks; the response to the reviewers is the guide to what changed.

Data and code

- The data files described in the data statement are deposited at Zenodo under CC BY 4.0: DOI 10.5281/zenodo.23202245.

- The analysis scripts, the decision rules that were written down before the runs, and the result files are in the public repository <https://github.com/neimasilk/volcarch-repo> (folders experiments/E227 to E231; MIT license).
- Both name the authors, which is why the article file and the response withhold them. The first reviewer asked for the release of the forms; please pass the DOI on if you see fit.

Other matters

- **Charges.** We note the Managing Editor's answer that there are none for standard publication; we take no paid open-access option.
- **Artificial intelligence.** The declaration on the use of artificial intelligence has been rewritten: it now names each stage at which the assistance was used, among them the re-derivation that found our errors, the analyses added for the revision, the locating and reading of the sources, and the drafting of the text.

With best regards,

Mukhlis Amien (corresponding author) and Go Frendi Gunawan
Universitas Bhinneka Nusantara

Response to the reviewers — manuscript OL-03-2026-11

Manuscript OL-03-2026-11, now titled *Uncoded basic vocabulary in six Sulawesi word lists of the Austronesian Basic Vocabulary Database*.

We are grateful to both reviewers. The first reviewer's comments corrected our classification and terminology and sent us to the literature on Makasar and Tolaki; the second reviewer's highlights told us, point by point, where a reader of the journal would be lost in Sections 2 and 3. We have tried to answer every one of them.

The revision also corrects errors of our own, which we found when we re-derived every number of the submitted text from the raw ABVD files and of which we told the editors before revising. Part C lists them. They change the abstract, the title and the claims of the paper: the classifier is now presented as a ranking aid rather than a detector of substrate vocabulary; the “two-method consensus”, the clustering, the ablation table and the section on Javanese script are removed, and the application to sixteen further lists is reduced to one paragraph without a geographic claim; the negative result about a shared layer is replaced by a statement of what a permutation test measures; and two subsections on Makasar and Tolaki answer the first reviewer's questions with counts from the data. One label definition is used throughout, so the Tolaki list released for inspection has 134 items rather than the 114 of the submitted Table 1.

The answers are given as bullets, as the Managing Editor advised: one for every point of the two reports (Parts A and B, in the order of the reports; the labels R1-1 to R1-13 and R2-1 to R2-24 are ours), with more detail where a point is substantive; Part C lists the corrections by the authors. Section, table and figure numbers are those of the revised manuscript unless “submitted” is said; the bold headings give the reviewer's point and so refer to the submitted version. This document carries no author names. The DOI of the data release and the address of the code repository name the authors; they are given to the editors in the cover letter and are withheld here, as they are in the article file.

Part A. Reviewer 1

- **Opening remark — the computational terminology is hard to follow; the caption “SHAP beeswarm plot for Model B” is the example given.** We agree. The second reviewer's highlights concern the same terms and are answered in Part B; for the figure behind that caption:
 - Figure 1 is now a plain bar chart of the mean absolute contribution of each input to the predictions of the full model, with the inputs named by what the code computes and, for each input, an arrow for its direction or a dot where there is no clear direction, explained in the legend; the software title, the raw variable names and the caption's sign error are gone.
 - Section 2.4 says what the bars and the arrows measure, and Section 3.5 says what the figure is and is not: a description of the classifier, not of the vocabulary.
- **R1-1 — “resist reconstruction to any proto-form” is too strong.** Accepted.
 - The wording is now “cannot be assigned to a higher-level proto-form” (first sentence of Section 1); the other sentences that carried the old wording, in the abstract and in the introduction, are gone with the rewriting.
 - The point reaches further than wording: Section 3.3 reports that 70.9 percent of Tolaki's uncoded forms have a same-meaning look-alike in Proto-Bungku-Tolaki or in one of 42 other Bungku-Tolaki lists (10.5 percent expected by chance; 95.9 percent for the coded Tolaki forms). The text states that this is a mechanical screen across lists that are almost all one compiler's, not a set of etymologies, and that recurrence within a subgroup says

nothing about origin — the reviewer’s own example of a word borrowed by the speakers of a low-level proto-language is given as the reason.

- **R1-2 — Makasar: how does the published 62 percent “open to investigation” fit our figure, and what does the study add?** The new Section 3.2 and Table 2 answer this.
 - The Introduction now gives the literature on Makasar that the reviewer names: its divergence within South Sulawesi (Mills 1975:491; Sirk 1989:71), the substratum raised and left open (Bulbeck 1992), and the 62 percent of Bulbeck, Pasqua and Di Lello (2000:103).
 - From ABVD, per meaning (201 meanings with a PMP entry), the Makasar form is in the PMP entry’s cognate set for 39.3 percent of meanings (Wilson 95 percent interval 32.8 to 46.2), coded in another set for 25.9 percent, and uncoded for 34.8 percent; Bugis 48.3 / 28.4 / 23.4, Sa’dan Toraja 48.0 / 33.0 / 19.0. We name the first class by what is counted (a form in the PMP entry’s set): it is a lower bound on the share that ABVD would count as retained, not retention itself, since an uncoded meaning cannot share a set with the PMP entry whatever its origin.
 - What distinguishes Makasar from its relatives in these data is one fact — more uncoded meanings (exact McNemar $p = 0.006$ against Bugis, 0.0001 against Sa’dan Toraja). Its lower share in the PMP set and its smaller share of common sets with each relative are that fact counted again, because on the meanings coded in both lists the shares are equal (70 against 69 of 110; 70 against 71 of 115).
 - A look-alike screen finds a PMP look-alike for 2 of 75 uncoded Makasar forms (about one expected by chance; the screen’s sensitivity on forms ABVD itself places in the PMP set is 38 percent, an upper bound for cognates the coders missed) and a Bugis or Sa’dan Toraja look-alike for 10 of 80 (chance 3.8 percent; 90 of 136 among coded forms). The screen therefore does not favour the reading that the uncoded forms are merely uncoded cognates; it leaves open what they are.
 - Section 3.2 traces the published figure from Bulbeck, Pasqua and Di Lello (2000:103) back to Bellwood (1997:115). The 38 percent is Blust’s lexicostatistical count against a 200-item PMP list as reported by Bellwood, beside a Western Malayo-Polynesian mean of 41 percent; ABVD’s PMP list is Blust’s too, so our 39.3 percent is a count of the same kind in the same tradition on another Makasar word list. The manuscript says that their agreement is not independent confirmation and that in its source the figure is not presented as low. Sirk’s Table 1 (45 / 42 / 60 for Makasar–Bugis, Makasar–Sa’dan, Bugis–Sa’dan) is given as a more independent comparison beside ABVD’s 41.4 / 39.2 / 53.1.
 - The 62 percent “open to investigation” is addressed in Section 3.2 as the complement of a retention count: it contains forms with cognates outside the PMP set as well as forms with none, and so cannot be set against the uncoded share of Table 1.
 - The manuscript states that the data are compatible with the lexical divergence reported in the literature, that they do not establish it, and that they say nothing about its cause. The 70 uncoded Makasar meanings, with the Bugis and Sa’dan Toraja forms and the PMP entry, are in the released files for anyone able to judge them.
 - Three smaller points. The report gives our earlier figure as 30.1 percent; the submitted Table 1 printed 30.9 percent. Mills’s statement of the divergence is on p. 491 of the dissertation, with its chart on p. 492 (the pages cited in the report hold cognate sets), and the text cites p. 491. We cite Bulbeck, Pasqua and Di Lello with the year 2000, the volume year of *Asian Perspectives* 39, and van den Berg’s paper on the demise of focus with 1996; the report has 2001 and 1989.

- **R1-3 — the cited source lists eleven subgroups; Sneddon 1993 could be cited.** Accepted. Section 1 now says that Blust (2013:82) lists eleven microgroups and that Sneddon (1993:2) counts nine plus Banggai as a group of one; the word “primary subgroups” is dropped, since neither source uses it, and the edition cited is the one consulted (2013).
- **R1-4 — Celebic is a supergroup.** Accepted. The sentence that named Celebic beside South Sulawesi and Muna-Buton is gone. Section 1 now names the four groups that our six lists represent (South Sulawesi, Muna-Buton, Wotu-Wolio and Bungku-Tolaki) and describes Celebic as a supergroup proposed by van den Berg (1996) and developed by Mead (2003), in whose version South Sulawesi lies outside it and some branches are called tentative. Because three of our four groups fall inside it, Section 3.7 says that the four are not four independent witnesses.
- **R1-5 — “parallel innovation” is a technical term.** Accepted; the term is gone, as is the subsection it headed.
 - We have not adopted “independent innovations” in its place, because the test behind the submitted negative result did not test that hypothesis (it compared a mean over twenty meanings with single draws of mostly coded vocabulary).
 - The permutation test that replaces it (Section 3.7) finds that, among uncoded forms, same-meaning forms are slightly more alike across lists than different-meaning forms (none of 10,000 permutations; four look-alike pairs against 0.5 expected; the coded forms, run as a positive control, show 472 against 8.5). The test asks only about the uncoded forms: shared vocabulary that ABVD’s editors recognised carries a set number and sits in the coded class, which the test does not examine beyond that control. The paper therefore makes no claim about a shared layer, positive or negative.
- **R1-6 — “three subgroups” but four listed; grouping; spelling; abbreviation.** Accepted on every point. The four subgroups are given as the reviewer gives them: Muna-Buton (Muna), Wotu-Wolio (Wolio), South Sulawesi (Bugis, Makasar, Sa’dan Toraja) and Bungku-Tolaki (Tolaki); Sa’dan Toraja is nowhere Celebic. We use *Makasar* and *Sa’dan Toraja*, and Section 2.1 says which ABVD list the Sa’dan Toraja data come from (list 226, “Tae’ (S. Toraja)”, from van der Veen 1940). The abbreviation of Bolaang Mongondow disappears with the submitted table and figure of the sixteen further languages (Part C).
- **R1-7 — “under-documentation” is unclear.** The word is withdrawn (Section 3.3).
 - What we meant is that 75 of the 209 Tolaki forms (35.9 percent) carry a cognate-set number. The list is not short of data: it fills 200 of 210 meanings, and 88.4 percent of its uncoded forms whose meaning the five Tolaki dialect lists cover recur in those lists.
 - What is sparse is the coding of the subgroup: by meaning, on shared meanings, the Tolaki list is coded for 38.3 percent, the dialect lists for 36.1 to 40.6 percent, and the 42 other Bungku-Tolaki lists for a median of 46.7 percent (five at or below the Tolaki figure); the unit matters because the share by form falls with the length of a list.
- **R1-8 — Muna *dh* is missing from the digraph list.** Two Muna forms contain *dh*: *akaradhaa* ‘to work’ and *idho* ‘green’. Both are cognate-coded and both were converted in the test; the sentence omitted *dh* (and Wolio *gh*, which occurs in no form). The count of converted forms (75: Muna 54, Tolaki 20, Sa’dan Toraja 1) is unaffected. No loanword was removed from the classifier’s data; *akaradhaa* carries ABVD’s loan flag and stays in. The script’s single-character placeholders for *gh*, *bh* and *dh* say nothing about Muna phonology, and the revision no longer gives phonetic values for the digraphs.

- **R1-9 — the glottal stop is written in several ways; same result under every convention?** Not quite, and the paper now says so (Section 3.6, Table 8).
 - The classifier sees only a written mark (ʔ or an apostrophe); the lists write it at very different rates (Makasar 36 percent of forms, Sa'dan Toraja 22, Bugis 21, Tolaki 14 — all with an apostrophe, and all 29 of them uncoded — Wolio four forms, Muna none), and whether these differences are notation or phonology is left to specialists.
 - Re-coding the mark as a consonant letter costs 0.016 of cross-validated AUC (0.020 to 0.024 for the held-out lists), leaving it unwritten 0.006, writing the pre-glottalised consonant as a geminate 0.007; unifying the apostrophe to ʔ changes no input. The AUC stays at or above 0.710 in every variant (held-out lists down to 0.677). By the rule written before the run, the result depends partly on the convention. The abstract's sentence that the profile is "robust ... rather than orthographic artifacts" is withdrawn.
 - On the *k* of some South Sulawesi sources: eleven Sa'dan Toraja forms end in *k* beside 45 that end in a glottal mark in the same source, and since ABVD's PMP entries for two of those meanings end in *k*, we do not treat that *k* as a glottal stop; the text says so. Glottal mark plus consonant occurs in 35 forms and is also counted by the cluster input.
- **R1-10 — a semantic feature cannot belong to a phonological fingerprint.** Accepted, and the point goes further than we had seen.
 - Of the 26 inputs of the submitted headline model (the one behind the AUC of 0.763, after the coverage variable had been taken out), 17 came from the written form, 8 from the meaning and one was the identity of the source list. That input was the strongest of the 26. The submitted text did not show this: its figure and its ranking were those of the 27-input model, in which the coverage variable came first.
 - The model of the revision has no list input; its AUC is 0.727 with form and meaning inputs and 0.672 with the form inputs alone (meaning alone 0.646; list identity alone 0.680). "Trained exclusively on phonological features" and "phonological fingerprint" are withdrawn. The term is now *profile*, defined in Section 1 as part orthographic and part semantic, and the title has changed accordingly.
- **R1-11 — example lexemes for each quadrant.** Table 7 gives two forms for each of the four cells of Table 6, printed as ABVD records them with the cognate-set number, the PMP entry of the meaning and the nearest same-meaning form in another Sulawesi list by the screen. The text states that they have not been assessed etymologically and that, being the forms the classifier scores most extremely, they illustrate its inputs by construction. The full pool of 92 example rows (73 forms) and the scores of all 1,357 forms are in the released files.
- **R1-12 — "false positive" is used in two senses; and is the first of them not a false negative?** Yes: with "positive" meaning Austronesian, an inherited form left in the residual is a false negative, and the submitted text called it a false positive throughout.
 - The confusion came from our two label sets and from a "rescue" step that was in fact a concept filter (Part C). The revision uses one label with two values, *coded* and *uncoded*; "positive", "false positive" and "unlabeled positive" are no longer used as names of classes (the word "positive" remains only in "positive control" and in "positive or negative").
 - We have taken up the reviewer's remark that loanwords are commonly adapted to the borrower's phonology (Blust 2012:556 is cited for it): Section 4 says that an uncoded form that looks like the coded vocabulary is not thereby inherited and deserves a specialist's attention as much as one that does not.

- **R1-13 — what “documentation gaps” means for Tolaki; release the residual items.**
 - - (a) We meant the second reading, cognates that exist but are not assigned in the database, and the phrase was badly chosen. On the first reading the data say only this: the list fills 200 of the 210 meanings, and of its 86 uncoded forms whose meaning the five Tolaki dialect lists also cover, 88.4 percent recur in those lists; whether a mainstream synonym exists that no list records, the study cannot show. The second reading is visible for part of the forms: of the 34 uncoded Tolaki forms that match an entry of the Proto-Bungku-Tolaki list, seven match an entry that ABVD does assign to a cognate set, and twelve of the 128 uncoded forms whose meaning has a PMP entry resemble that entry by the screen (about two expected by chance; one is loan-flagged). For the rest the study cannot decide between a coding gap, a chance resemblance and a loan, and Section 3.3 says so.
 - - (b) All forms are released, not only Tolaki’s: one file with the 1,357 forms (ABVD identifiers, forms as recorded, cognate numbers and loan flags, the label, three out-of-sample scores, the nearest look-alikes), one with the 438 uncoded forms, and three smaller files (the 70 uncoded Makasar meanings with their relatives’ forms, the twelve Tolaki look-alikes of PMP entries, the example pool behind Table 7), with a column description, deposited at Zenodo under CC BY 4.0. The DOI is given to the editors in the cover letter and withheld here because the record names the authors. The Tolaki subset has 134 items rather than the 114 of the submitted Table 1 because one label definition is now used (Part C).

Part B. Reviewer 2

We have rewritten Sections 2 and 3 for readers who do not work with classifiers: the terms the reviewer marked are explained in words, and the internal experiment codes are gone. The bullets follow the 24 highlights in the order of the pages of the reviewer’s copy.

- **R2-1 — a reference for the claim that vocabulary “resists reconstruction”.** We found no source that states the claim as we had written it, nor one that states it for Austronesian languages in general. The sentence now says only that in many Austronesian languages some forms cannot be assigned to a higher-level proto-form; it has no reference of its own, and the sentences that follow give what the literature does state: Reid (1994), who counted forms with no known cognate in three Philippine Negrito languages, and the literature on Makasar. The stronger triple claim of the submitted version (no cognates, non-canonical structure, irregular correspondences) is dropped.
- **R2-2 — what are E027 / E028 / E022?** Internal experiment codes; they appeared 25 times in the text and in the titles of two figures. They are gone from the text, which speaks of the label, the classifier, the screen and the permutation test. (The data statement names the folders of the code repository that hold the scripts, “experiments/E227 to E231”; these are folder names, not labels used in the text.)
- **R2-3 — the source of the semantic domains.** The seven domains and the “core” set are groupings made by the authors for this study, not Concepticon or the World Loanword Database; Section 2.3 says so, and it no longer calls the core set “Swadesh-100” (it covers 174 of the 210 meanings).

- **R2-4 — the “language identity” feature.** It was an integer numbering the six lists, and in the submitted headline model (26 inputs) it was the strongest input (see R1-10). It has been removed, and Section 2.3 says so.
- **R2-5 — what the classifier is and what “robustness” means.** Section 2.4 now opens with a sentence that says in plain words what a classifier is, and goes on to say what its score means; “robust” is no longer used. Three classifiers give the same picture (Table 3), which is all the submitted “robustness” amounted to.
- **R2-6 — implemented in scikit-learn / XGBoost — Python?** Yes; Section 2.4 gives the language and the version of each library.
- **R2-7 — what the false positives stand for.** The term is no longer used (see R1-12).
- **R2-8 — the “E022 binary label” cannot be found in the rule-based section.** Because the label the models used was not the one that section described. The revision has one label, defined in Section 1 and, with what follows from it, in Section 2.2.
- **R2-9 — kappa: agreement?** Section 3.5 defines it where Table 6 reports it (Cohen’s kappa between the label and the model’s out-of-sample score at a threshold of 0.5; 0.31). The submitted 0.61 was computed on the training data and is withdrawn.
- **R2-10 — why k = 5 to 30?** There was no reason; the clustering is removed (Part C).
- **R2-11 — “both models” — which?** The first of them (31 inputs, four of which encoded the label) is removed; one model family remains, with a form-only variant beside it.
- **R2-12 — “near-perfect” and how to read Table 2.** The near-perfect figure belonged to the removed model. The new Table 3 gives, for each classifier, the AUC, the accuracy beside the accuracy of calling every form coded (0.719 against 0.677), and the precision, recall and F1 of the uncoded class (the submitted F1 of 0.82 was the coded class’s). Section 2.5 explains each column in words, and Section 3.4 reads the first row of the table.
- **R2-13 — AUC: what does it stand for and mean?** Defined in Section 2.5: the probability that a randomly chosen uncoded form outranks a randomly chosen coded one; 0.5 is chance, 1.0 is perfect.
- **R2-14 — which range counts as moderate or reliable?** Section 2.5 says that verbal scales exist and gives one (Nahm 2022, Table 4), but that they are conventions without an empirical basis (White et al. 2023), and the paper reports numbers rather than labels. The “0.65 threshold” of the submitted text was our own working line and is dropped.
- **R2-15 — what “generalises across Sulawesi languages” means.** Section 2.5 defines it (a model trained on five lists ranks the forms of the sixth) and Section 3.4 reports it: AUC 0.614 to 0.809 for the held-out list, mean 0.701, and — new — that at a threshold of 0.5 the classifier is below the majority answer for five of the six lists. The ranking carries over to an unseen list; the classification does not.
- **R2-16 — “feature ablation” and the variable in typewriter font.** The word “ablation” is no longer used; where the classifier is run again without a group of inputs, the text says so in those words (Sections 2.6 and 3.6). The variable in question no longer exists — its values in the code were stale constants, not the coverage they were named for — and nothing is printed in typewriter font; where the code’s own name for an input says more than the input measures, Section 2.3 says so.
- **R2-17 — SHAP.** Section 2.4 explains what the bars and the arrows of Figure 1 measure; Section 3.5 says that the figure describes the classifier and not the vocabulary.
- **R2-18 — CV.** Cross-validation is explained in words in Section 2.5 (five parts, ten repetitions, fifty test sets).
- **R2-19 — the symbol Δ .** It now occurs only as a column heading of Table 8, and the note to the table says what it is: the change in AUC from the first row.

- **R2-20 — the panels of Figure 2 are neither referred to nor explained.** The submitted Figure 2 is removed; Table 6 replaces it, with examples in Table 7. (Figure 2 of the revision is the new workflow diagram.)
- **R2-21 — why ‘One Hundred’, ‘Fifty’, ‘Twenty’, ‘to stand’ and ‘to hit’?** Section 3.5 answers for four of the five: ‘Fifty’ and ‘Twenty’ appear to be decimal compounds whose parts recur across the lists (uncoded in four, coded in Bugis); ‘to stand’ and ‘to hit’ were on our earlier list of inherited meanings, and the “consensus” had been computed on a label that ignored that list. ‘One Hundred’ (uncoded in four lists, coded in Bugis) is a different case: its forms are not uniform across the lists, the text says only that it does not belong with the compounds, and we give no account of it; the forms are in the released file. Out of sample, no meaning is “uncoded, profile” in four or more lists.
- **R2-22 — DBSCAN parameters.** Removed with the clustering.
- **R2-23 — Table 5 is cut off.** The submitted Table 5 and Figure 4 are removed; Section 3.8 reports the sixteen-list application in one paragraph without a geographic claim (Part C).
- **R2-24 — define “fingerprint” at the start.** The term is now *profile*, defined in Section 1; because its inputs include the meaning, it is not called phonological.

We have kept the three things the reviewer singled out for approval: the limitations (a paragraph of Section 4, rewritten for what the revision claims; limits are also stated with each result), the declaration on the use of artificial intelligence (rewritten to name each stage at which the assistance was used), and the open code and data (now with a Zenodo deposit). The reviewer also found the Discussion easier to follow than Section 3, and its insights of potential interest. Much of the submitted Discussion rested on results that did not survive our re-derivation (Part C) and is withdrawn; the revised Section 4 is shorter and says what the measurements do and do not show for the profile, for Makasar and for Tolaki.

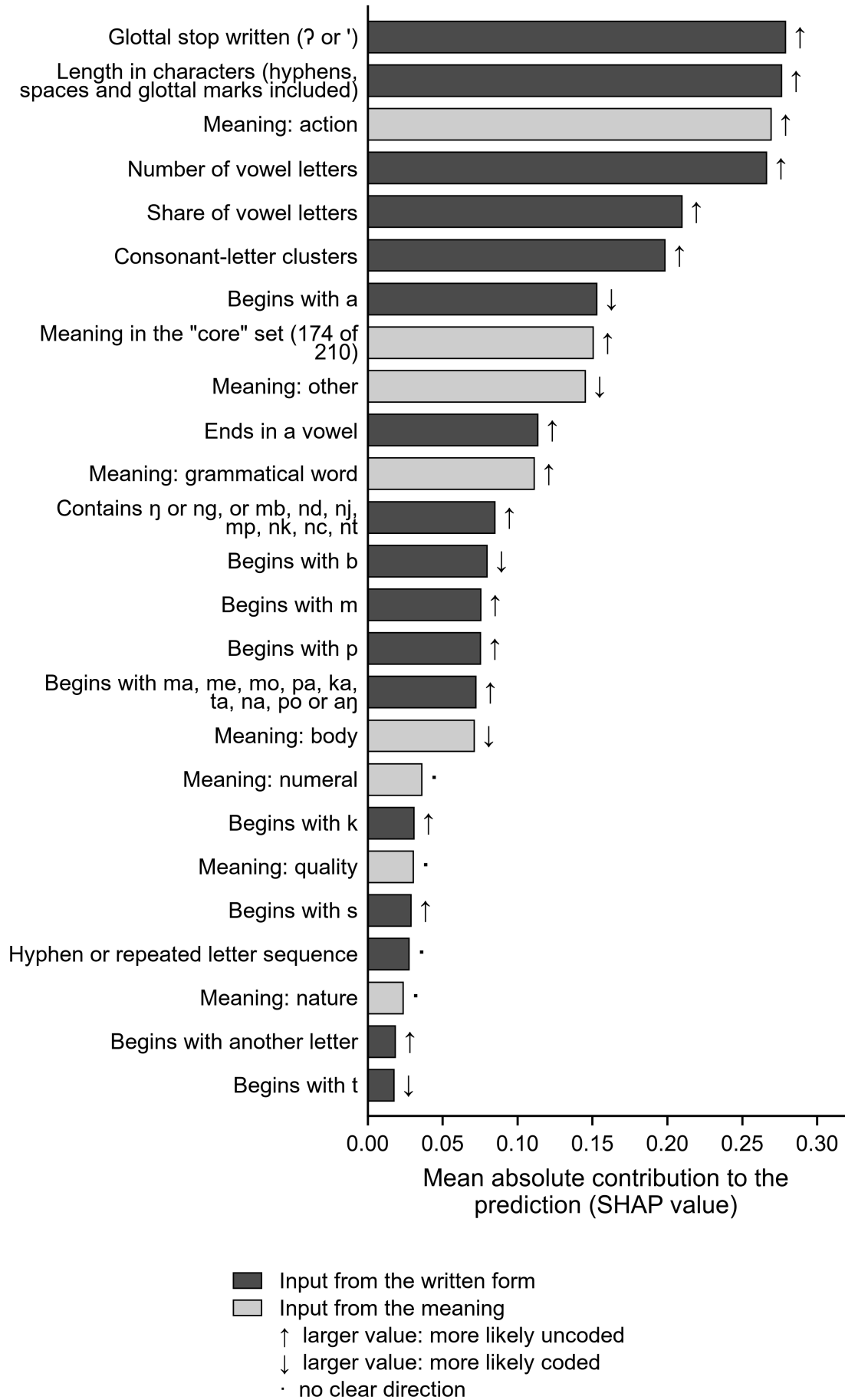
Part C. Corrections by the authors

Found when every number of the submitted text was re-derived from the raw files before this revision; not raised as such by the reviewers, though several of their questions lead to them. The stored result files reproduce from the data; the manuscript text did not describe them correctly.

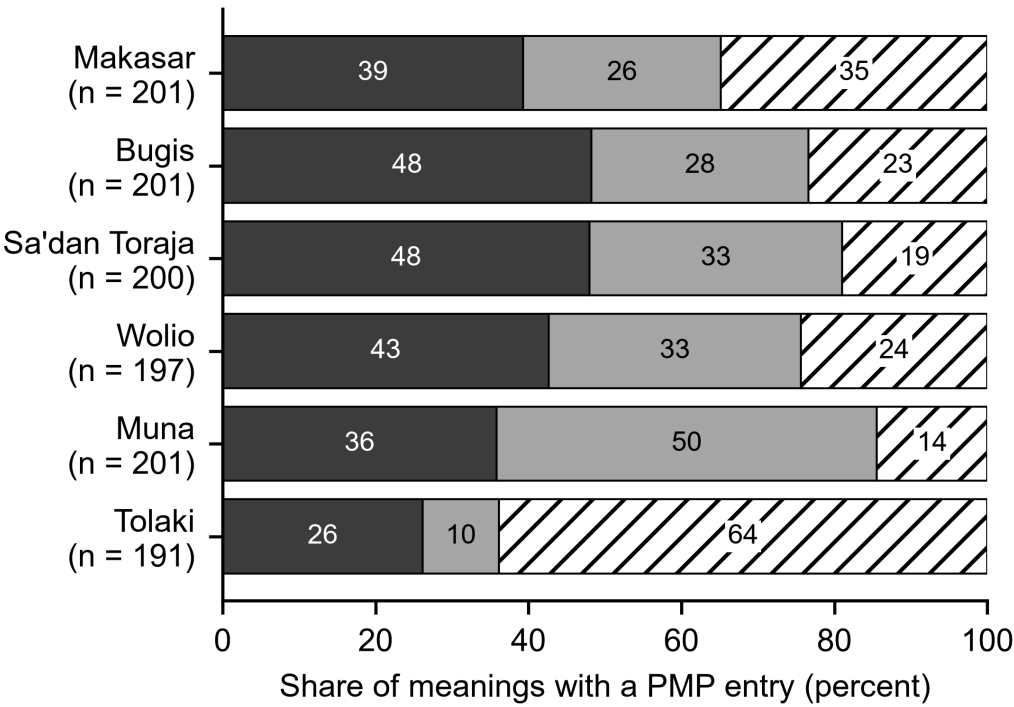
- **Two residual sets presented as one.** Submitted: 356 forms in Table 1; 438 in every model and in the abstract as “26.5 %”. Revised: one label: 438 uncoded forms = 32.3 percent; per list 34 / 62 / 80 / 83 / 45 / 134 (Table 1).
- **“Proto-Austronesian cross-check”.** Submitted: described as a comparison of residual forms with reconstructions. Revised: it compared no forms (all uncoded forms for 15 meanings were removed); step deleted.
- **Loanword filter.** Submitted: Sanskrit, Arabic and Malay trade-word patterns. Revised: matched five unrelated words; deleted.
- **“Fewer canonical Austronesian prefixes”.** Submitted: abstract and three places in the text. Revised: the onset-string input is more frequent among uncoded forms (37.0 vs 25.2 %) but not separable from length; neither “fewer” nor “more” is claimed (Section 3.5).
- **Headline AUC 0.763 “phonological only”.** Submitted: 26 inputs including the identity of the source list. Revised: form + meaning 0.727 (held-out list 0.701); form only 0.672 (0.641); no list input.
- **“Two-method consensus”, $\kappa = 0.61$, 266 forms.** Submitted: agreement computed on the training data. Revised: out of sample $\kappa = 0.31$; cells 172 / 266 / 105 / 814 (Table 6); the consensus framing removed. (That 266 recurs is a coincidence: on the training data 266 of the 438 uncoded forms scored 0.5 or more; out of sample 172 do and 266 do not.)

- **p = 0.569; “optimal k = 30”.** Submitted: a test that did not test the hypothesis; the edge of the search range. Revised: permutation test: a small excess of same-meaning similarity among uncoded forms (Section 3.7); clustering removed; no claim about a shared layer either way.
- **Sixteen-language application.** Submitted: mean score 0.606 vs 0.393 “significantly”; Table 5 with four stale constants. Revised: two list-level inputs drove the pattern; without them mean AUC 0.638 and no geographic claim (Section 3.8); the submitted Table 5 and Figure 4 removed.
- **Robustness section.** Submitted: 0.772 against 0.774 and similar pairs; “does not depend on form length at all”. Revised: the submitted numbers were not those of the model that section named: they came from the 26-input model under another division into folds. For the models of the revision (Section 3.6), digraph conversion changes nothing (75 forms) and removing both size inputs costs 0.014 / 0.023 AUC; the sentence is withdrawn.
- **Figure 1 caption; Table 2 F1; class weighting.** Submitted: sign reversed; F1 0.82 = coded class; XGBoost described as adjusted for class imbalance. Revised: Figure 1 redrawn; the F1 of the uncoded class is 0.480, and the accuracy is 0.719 beside 0.677 for calling every form coded (Table 3); XGBoost was run without class weighting, and Section 2.4 says so.
- **“Swadesh-100” flag; coverage variable; “syllable”.** Submitted: a flag that covers 174 of the 210 meanings; a variable that held stale constants; “syllables” that were groups of vowel letters. Revised: the core set is described as what it is (Section 2.3); the coverage variable and the syllable count are no longer used.
- **The submitted Section 4.5 (Javanese script).** Submitted: “convergent evidence”. Revised: removed: not asked for, outside Sulawesi, resting on properties that changed, and quoting a percentage for the wrong group.
- **References.** Submitted: Ross 2005 given as an article in this journal (it is a book chapter); two entries in our bibliography file for works that do not exist (neither was cited); seven further entries with a wrong entry type, publisher or pages. Revised: every entry of the reference list was checked against the publication, its catalogue record or, for a few page ranges, the reference list of another work; one page range that could not be confirmed is omitted; Ross 2005 is no longer cited and the two entries are deleted.

Gambar 1 (digambar ulang: legenda)



Gambar 3 (digambar ulang: angka di segmen berarsir)



■ in the PMP entry's cognate set ■ coded in another set ▨ no cognate set

Gambar 4 (digambar ulang: legenda dan kata 'chance')

